Privacy-preserving Speech Emotion Recognition through Semi-Supervised Federated Learning

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01 Context & approach

The challenge

Speech emotion models need costly annotations, and central training requires collecting sensitive recordings. Federated self-training uses each device's unlabeled speech alongside its small labeled subset, while exchanging model updates.

Our contribution

Combine spectral, temporal and channel attention in a speech CNN.
Filter local pseudo-labels using a device-specific confidence schedule.
Compare four speaker/client splits at the same 10% label budget.

1 Represent emotional speech

Convert local audio to log-mel features; an attention CNN focuses on spectral, temporal and channel cues.

2 Learn from unlabeled clips

Train jointly on true labels and confident pseudo-labels; filtering accounts for training progress and each client's participation.

3 Aggregate model updates

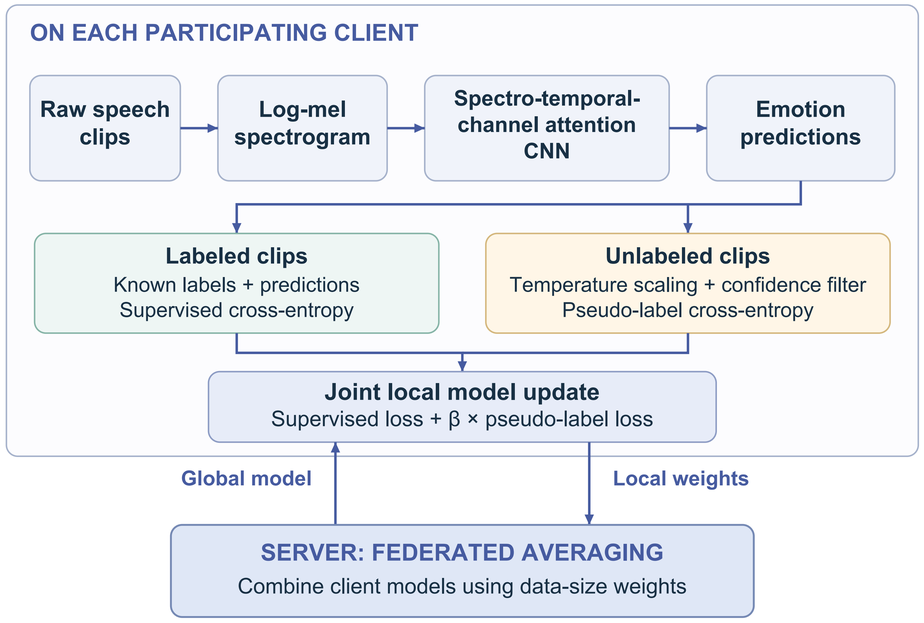
FedAvg combines local parameters. Raw speech stays on clients; no server training dataset is required.

Scope

One four-emotion corpus; random/per-speaker splits also differ in client count, and results omit dispersion. Raw-data locality does not provide differential privacy or protection against model-update attacks.

02 Method

Federated self-training improves speech emotion recognition when only 10% of recordings are labeled.



Core formulation

$$\mathcal{L}_k = \mathcal{L}_s(\mathcal{D}_L^k) + \beta \mathcal{L}_u(\mathcal{D}_U^k)$$

Client k combines supervised loss on labeled speech D_L^k with loss on confident pseudo-labels from D_U^k; β sets their relative influence.

Paper

Code

03 Results

10%
labeled speech
Remaining clips used unlabeled

56.83%
unweighted accuracy
Unseen speakers; per-speaker clients

Unlabeled speech helps at a fixed label budget

Speaker/client split	Supervised 10%	Self-training 10% (Ours)	Supervised 100%
Unseen / random	46.72	**53.94**	60.52
Unseen / per speaker	46.55	**56.83**	59.55
Overlap / random	51.58	**59.04**	64.78
Overlap / per speaker	49.97	**59.72**	62.79

IEMOCAP improvised speech, four emotions; mean unweighted accuracy (%) over five runs per split. Random/per-speaker partitions use 10/8 clients.

Fully labeled reference models

Same architecture	Unseen speakers (%)	Overlapping speakers (%)
Centralized	62.58	69.04
Federated	60.52	64.78

100% labels, random partitions; mean unweighted accuracy. Federated setup: 10 clients, 100 rounds, 80% participation, one local step.

Takeaway

Self-training improves every 10%-label comparison, while remaining below the corresponding fully labeled reference. It keeps **raw recordings local**; the experiments establish no formal privacy guarantee.